BASiS: Transferring Dialogue Progression Strategies to Large Language Models through Bayesian Dialogue Modeling

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We propose BASiS (Bayesian Style-based Selection), a framework for scaling the interactional knowledge encoded in a small, high-quality dialogue corpus to large-scale LLM training. We focus on settings where dialogue must progress according to the user's state and the stage of the interaction as exemplified in emotional support, education, and healthcare. In such settings, dialogue quality depends not only on utterance content but also on how the dialogue develops across turns, which involves procedural interactional knowledge of how to adapt response strategies as the interaction evolves. Although such knowledge is implicitly encoded in small, high-quality corpora, these corpora alone provide limited data for LLM training. To overcome this challenge, BASiS models the progression patterns in the high-quality corpus and uses the resulting model to guide training-data selection from a large general-purpose corpus. It extracts dialogue states, response strategies, and state transitions from the small corpus and represents the probabilistic relationships as a Bayesian dialogue model. The model scores candidate responses in a large corpus according to how well they support the dialogue progression exhibited in the high-quality corpus, and the scores are used to construct preference pairs for LLM fine-tuning with Direct Preference Optimization (DPO). We evaluated BASiS in emotional support, mathematics tutoring, and medical history-taking against the base model and a model trained on randomly selected in-category data. In the human evaluation, BASiS achieved the highest mean score on all 19 evaluation dimensions and was selected most often in all three domains. In the large-scale LLM-based evaluation, BASiS achieved the highest mean score on all 21 evaluation dimensions, with broad statistically significant improvements in emotional support and mathematics tutoring.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; **Discourse, dialogue and pragmatics**; Learning from implicit feedback; • **Human-centered computing** → *Natural language interfaces.*

Additional Key Words and Phrases: dialogue progression, data selection, Bayesian dialogue modeling, response strategy, preference optimization, emotional support dialogue, large language models

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1 Introduction

Advances in instruction tuning and reinforcement learning from human feedback (RLHF) have enabled recent large language models (LLMs) to follow instructions, generate accurate responses, and sustain natural and coherent multi-turn conversations across a wide range of topics [10]. In settings such as emotional support, mathematics tutoring, and medical history-taking, however, a system must do more than provide correct information: it must strategically progress the dialogue flow according to the interlocutor's state and the current stage of the interaction [6, 12]. In mathematics tutoring, for example, the system should not simply provide the correct answer; it must first understand

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the learner's current reasoning and then choose a question, hint, or explanation that helps the learner take the next step independently. Producing such responses requires the system to control dialogue progression across turns.

LLM-based response generation has approached this requirement in two main ways. Guideline-based methods describe desirable response strategies in natural-language dialogue guidelines and guide generation to follow them [3, 16]. Data-driven methods instead adapt model behavior through instruction tuning or by selecting high-quality training examples [18, 21]. However, the former requires dialogue guidelines to be designed manually for each setting, making it difficult to enumerate implicit features such as response timing and multi-turn progression. The latter depends on sufficient training data, whereas detailed interactional knowledge is often encoded in small, carefully constructed dialogue corpora that alone provide too few examples for LLM training.

A separate line of research has modeled dialogue progression through dialogue actions, response strategies, and their transitions. Ganesh et al. predicted the next teacher talk move in classroom dialogue [2]; Wan et al. modeled relationships between user emotion states and response strategies as a heterogeneous graph [15]; and Rudolph et al. regularized next-dialogue-act prediction using a transition matrix derived from counseling conversations [13]. These studies capture important structure beyond surface utterances, but primarily evaluate action or strategy prediction. Their integration with LLMs for natural-language response generation, and evaluation of whether the modeled progression appears in generated responses, remain limited.

To bridge these two lines of work, we propose BASiS (Bayesian Style-based Selection), a method for transferring dialogue progression from a small, high-quality corpus to an LLM. BASiS first induces a progression model comprising conversational states, response strategies, state transitions, and their probabilistic relationships. It then scores candidate responses in a large general-purpose corpus according to their compatibility with that progression and constructs preference pairs from the scores. Finally, it adapts the LLM using Direct Preference Optimization (DPO) [11] with Low-Rank Adaptation (LoRA) [4]. This combination uses large-corpus selection to scale the interactional knowledge encoded in the small corpus, while reducing the need to manually author detailed dialogue guidelines. By maintaining a probability distribution over conversational states, BASiS can account for dialogue history and multiple valid response strategies. We evaluate the same pipeline in emotional support, mathematics tutoring, and medical history-taking by changing the reference corpus and candidate pool.

The remainder of this paper is organized as follows. Section 2 reviews related work on guideline-based response control, dialogue-history-based prediction of dialogue actions and response strategies, and training-data selection for LLMs. Section 3 describes the four steps of the BASiS pipeline. Section 4 presents the evaluation design, results, and discussion. Finally, Section 5 concludes the paper.

2 Related Work

Guideline-based approaches make desired dialogue behavior explicit and use it to control response generation. DialGuide represents each guideline as a natural-language condition specifying when it applies and an action specifying what the response should contain [3]. It retrieves guidelines relevant to the current dialogue context and uses them to generate or verify a response. GuidedTOD instead represents actions specified in operational guidelines as states in a Chained Prior, a Markov chain whose transition probabilities are calibrated from curated service data [16]. During inference, the Chained Prior is updated as the dialogue progresses and is used to rerank candidate outputs. These studies show that explicit dialogue guidelines can guide generation and selection according to the current stage of a dialogue. However, developers must write the guidelines and redesign them when the desired dialogue progression changes.

Dialogue progression can also be modeled from corpora. Ganesh et al. predict the teacher's next talk move from the preceding classroom dialogue and talk moves [2]. EmoDynamiX represents discourse-level relationships between fine-grained user emotion states and system response strategies as a heterogeneous graph [15]. Rudolph et al. estimate dialogue-act transition patterns from counseling conversations and use them to regularize next-dialogue-act prediction toward the observed dialogue flow [13]. Together, these studies establish that dialogue progression can be represented as corpus-derived states, actions, strategies, and transitions rather than only as manually written rules.

Data-centric approaches select training examples from a large candidate pool to adapt LLM behavior efficiently. LIMA emphasizes careful curation for data quality and diversity [21]. AlpaGasus filters instruction data using LLM-based quality scores, whereas DEITA combines instruction complexity, response quality, and diversity-aware selection [1, 7]. LESS estimates the influence of individual examples on downstream performance and selects those expected to be most useful [17]. These studies show that a criterion representing the desired behavior or capability can guide the construction of effective adaptation data.

Corpus-based dialogue modeling and data-centric adaptation together suggest a broader opportunity: using dialogue progression extracted from a corpus as a criterion for selecting LLM training data. In settings such as mathematics tutoring, the suitability of a response depends not only on its standalone quality or topical relevance, but also on the learner's current state, the preceding interaction, and the stage toward which the response moves the dialogue. The corpus-based models above use dialogue structure primarily to predict the next action or strategy rather than to supervise natural-language response generation. Conversely, the data-selection methods above assess examples by quality, complexity, diversity, or downstream influence without representing how each response advances a multi-turn interaction as learned from a high-quality dialogue corpus. BASiS addresses this gap by converting the implicit dialogue progression in such a corpus into a probabilistic selection criterion and then into preference data for LLM adaptation. In this way, BASiS enables corpus-derived interactional knowledge to shape LLM-generated responses at scale, without requiring developers to specify the complete dialogue progression through manually authored guidelines.

3 BASiS

3.1 Concept

We propose BASiS, which adapts an LLM to the response strategies and dialogue progression encoded in a small, high-quality corpus through Bayesian data selection. BASiS converts these interactional patterns into a probabilistic dialogue model. Instead of training directly on the small corpus, it uses the model as a criterion for selecting data from a large general-purpose corpus. BASiS evaluates each dialogue instance in the large corpus according to its compatibility with the target corpus's dialogue progression. It thereby incorporates the interactional knowledge in the small corpus into training data without requiring detailed dialogue guidelines to be written manually for each setting. The selected data are ultimately used for DPO training, which adapts the LLM to the response strategies and progression patterns of the small corpus.

3.2 BASiS Pipeline

Figure 1 presents the four steps of the BASiS pipeline, from modeling dialogue progression to selecting training data and adapting an LLM. The upper pathway A, *Dialogue Progression Modeling*, comprises Steps 1 and 2. Step 1 takes the dialogue text and existing annotations in a small, high-quality corpus as input and extracts corpus-specific conversational states, response strategies, state transitions, and probability estimates. Step 2 takes these outputs, normalizes the probability

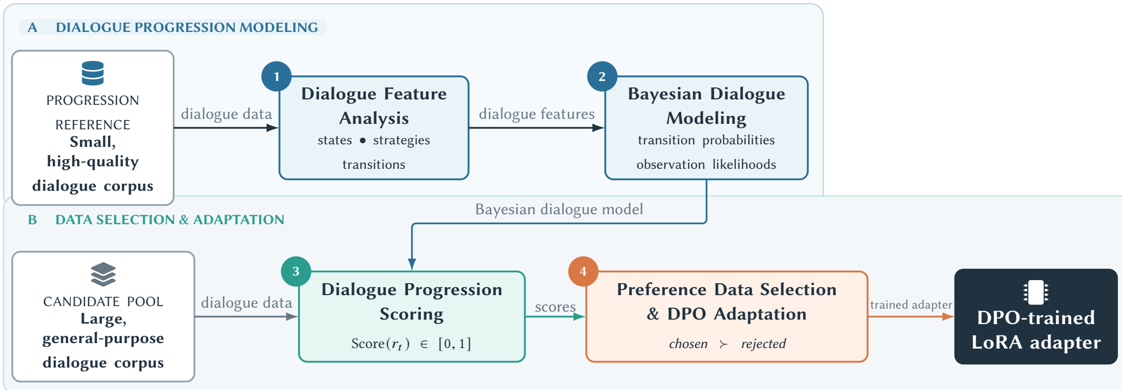

Fig. 1. BASiS pipeline. The upper pathway extracts dialogue features and probability estimates from a small, high-quality corpus and formalizes them as a Bayesian dialogue model. The lower pathway applies the fixed model to candidate responses from a large general-purpose corpus and uses the resulting preference data to train a LoRA adapter with DPO. The numbered circles correspond to the four steps described in the text.

estimates, and organizes them as a fixed Bayesian dialogue model. The lower pathway B, *Data Selection & Adaptation*, comprises Steps 3 and 4. Step 3 takes the fixed Bayesian model and dialogue data from a large general-purpose candidate pool as input and assigns each candidate response a progression-compatibility score between 0 and 1. Step 4 takes the candidate responses and their scores, constructs *chosen–rejected* preference pairs, and outputs a LoRA adapter trained with DPO. Applying this adapter to the base LLM yields a model adapted to the progression patterns in the reference corpus. Thus, the small corpus serves as the progression reference, whereas the large corpus provides the candidate training data. The same pipeline can be applied in other dialogue settings by changing these two corpora.

Step 1: Dialogue Feature Analysis

The input to Step 1 is the dialogue text and existing annotations for 80 conversations selected from a small, high-quality corpus. All 80 conversations are supplied together in a single prompt rather than divided across calls. The output is a representation of the corpus comprising conversational states, response strategies, state transitions, and probability estimates. For each dialogue turn, a *conversational state* describes the interlocutor's situation or the current stage of the interaction, a *response strategy* describes the observable communicative function of the response, and a *state transition* describes the change from the state before the response to the state after it.

The prompt instructs the LLM to consolidate semantically similar labels across the selected dialogues, define 4–7 conversational states, keep states conceptually distinct from response strategies, assign these labels to the turns, and identify which states represent desirable progression. Existing annotations provide evidence about the communicative role of each turn, but their labels are not mechanically copied as state names. Rather than calculating probabilities from label frequencies, the LLM outputs numerical estimates of the initial-state probabilities, state-transition probabilities, and observation likelihoods based on the corpus as a whole. We then check for missing labels, values outside $[0, 1]$, and rows of probabilities that do not sum to one; Step 2 normalizes the latter where necessary. For example, ESConv, the emotional-support dialogue corpus used in our evaluation [6], yielded `opening_rapport`, a state in which greetings and brief check-ins create a safe atmosphere, and `empathic_support`, a state in which the help seeker's emotions are acknowledged to alleviate isolation and distress. Its response strategies included `explore_question`, which asks

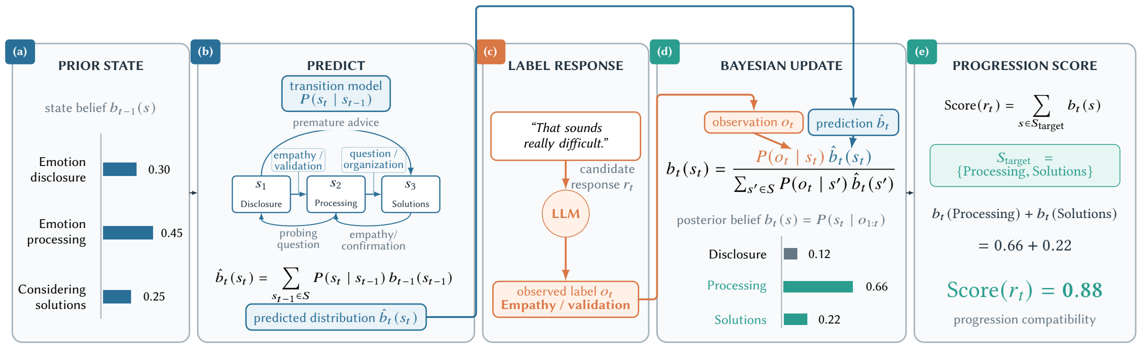

Fig. 2. Schematic example of candidate-response evaluation in Step 3. Panels (a)–(e) show the five internal operations: prior-state specification, prediction, response labeling, Bayesian update, and progression-compatibility scoring. The blue and orange paths show where the predicted distribution and observation label enter the update equation. Here, the probabilities of the target states Processing and Solutions are summed to obtain Score = 0.88.

contextually relevant questions to understand the seeker's situation, and `reflect_validate`, which paraphrases and validates the seeker's words and emotions.

Step 2: Bayesian Dialogue Modeling

Step 2 takes the labels and probability estimates produced in Step 1 as input and outputs a normalized, fixed Bayesian dialogue model representing the progression in the reference corpus. Let $S = \{s_1, \ldots, s_K\}$ denote the conversational states and $O = \{o_1, \ldots, o_L\}$ the observation labels corresponding to response strategies. The model consists of the initial-state probability $P(s_1)$, state-transition probability $P(s_t \mid s_{t-1})$, and observation likelihood $P(o_t \mid s_t)$. The transition probabilities represent the relative tendency, estimated from the selected reference dialogues, to move from s_{t-1} to s_t; the observation likelihood represents the probability of observing strategy o_t in state s_t. Step 2 normalizes the estimates from Step 1 into valid probability distributions and fixes them as the initial-state distribution, transition matrix, and observation-likelihood matrix. It performs no additional corpus analysis or model fitting. The resulting model represents probabilistic relationships between states and strategies rather than memorizing utterances, so it can be applied to candidate responses outside the small corpus.

Step 3: Dialogue Progression Scoring

Step 3 takes the fixed Bayesian dialogue model from Step 2 together with dialogue histories and candidate responses from the large general-purpose corpus as input. Without rebuilding or fitting the model, it applies Bayesian inference to each candidate and outputs a progression-compatibility score. Figure 2 expands this step into five internal operations: (a) specifying the prior state distribution, (b) predicting the next-state distribution, (c) labeling the candidate response, (d) updating the state distribution, and (e) computing the score.

To construct the state distribution for each candidate, we group the large corpus by conversation ID and process responses from the beginning of each conversation in the order indicated by the turn index. The first evaluated response starts from the initial-state distribution produced in Step 2. Each preceding assistant response is assigned one of the response-strategy labels and used to update the Bayesian state distribution; the resulting posterior is carried forward as the prior for the next response. The LLM receives the complete dialogue history from the beginning of the conversation when assigning a response label. When a *chosen* response and its generated *rejected* alternatives are compared for the same context, all candidates use the same state distribution immediately before that response.

Before a candidate response is presented, the dialogue state is maintained as a probability distribution over the state set S rather than fixed to a single state (panel (a) in Figure 2). In the example, the state distribution through turn $t-1$, $b_{t-1}(s)$, assigns 0.30 to Emotion disclosure, 0.45 to Emotion processing, and 0.25 to Considering solutions. The predicted state distribution at turn t is then computed using the fixed transition probabilities (panel (b)):

$$\hat{b}_t(s_t) = \sum_{s_{t-1} \in S} P(s_t \mid s_{t-1}) b_{t-1}(s_{t-1}). \tag{1}$$

The state-transition diagram in panel (b) of Figure 2 schematically shows transitions among Disclosure, Processing, and Solutions, together with representative response strategies observed around those transitions. Equation (1) maps the previous distribution b_{t-1} to the predicted distribution $\hat{b}_t$ using $P(s_t \mid s_{t-1})$, which summarizes the transition patterns in the reference corpus.

Next, an LLM maps candidate response r_t to exactly one observation label o_t from the fixed set of response-strategy labels (panel (c)). The input comprises the dialogue history, candidate response, and the labels with their descriptions; the LLM is not allowed to create a new label or state name. Only the selected observation label is used in the subsequent Bayesian computation. In the example, "That sounds really difficult." is labeled *Empathy / validation*. The LLM does not assign a conversational state directly. Instead, the fixed observation likelihood for o_t is used to update the predicted distribution by Bayes' rule:

$$b_t(s_t) = \frac{P(o_t \mid s_t)\hat{b}_t(s_t)}{\sum_{s' \in S} P(o_t \mid s')\hat{b}_t(s')}. \tag{2}$$

Here, $b_t(s_t)$ is the posterior probability $P(s_t \mid o_{1:t})$ conditioned on the observation history $o_{1:t}$ after incorporating the observations into the predicted distribution $\hat{b}_t(s_t)$. In panel (d) of Figure 2, the orange path shows that observation o_t determines the likelihood term $P(o_t \mid s_t)$, whereas the blue path shows where prediction $\hat{b}_t(s_t)$ enters the update. The denominator normalizes the posterior over all states. In this example, the updated distribution assigns 0.12 to Disclosure, 0.66 to Processing, and 0.22 to Solutions.

Finally, let $S_{\text{target}} \subseteq S$ be the set of states defined as desirable for the target dialogue progression. The progression-compatibility score of candidate response r_t is the total probability assigned to S_{target} in the updated state distribution:

$$\text{Score}(r_t) = \sum_{s \in S_{\text{target}}} b_t(s). \tag{3}$$

The progression-compatibility score ranges from 0 to 1. A larger value indicates that the state distribution after observing the candidate response is more strongly aligned with states considered desirable in the target corpus. In panel (e) of Figure 2, $S_{\text{target}} = \{\text{Processing}, \text{Solutions}\}$, so the score is $0.66 + 0.22 = 0.88$. More generally, S_{target} may include states such as processing emotions or considering appropriate solutions in emotional support, advancing the learner's own reasoning in mathematics tutoring, or progressively organizing necessary symptom information in medical history-taking.

This scoring procedure maintains a probability distribution over states instead of fixing a single response strategy as the correct next action. Even when each candidate response receives one observation label, responses associated with different strategies can each obtain a high progression-compatibility score if those strategies have high observation likelihoods in the desirable states. Thus, when several response strategies are valid for the same context, Step 3 can evaluate them without restricting the choice to a single strategy.

Step 4: Preference Data Selection and DPO Adaptation

Step 4 takes the candidate responses and progression-compatibility scores from Step 3 as input and outputs a LoRA adapter trained with DPO. Each original context contributes one actual assistant response rather than a pre-existing set of alternatives. We first score these responses and retain high-scoring ones as *chosen* candidates. For each retained context, an LLM generates eight natural responses that are weaker in their response strategy. We do not use responses taken from other dialogues as negative examples. All eight alternatives are scored from the same state distribution immediately before the response, and the alternative with the largest score difference from the chosen response is selected as *rejected*.

We retain a BASiS preference pair only when the chosen score is at least 0.70, the rejected score is at most 0.55, and their difference is at least 0.20. The resulting pairs are used for Direct Preference Optimization (DPO) [11], which optimizes the model to prefer chosen responses over rejected ones. Rather than updating all model parameters, we apply Low-Rank Adaptation (LoRA) [4] and update only the added low-rank parameters. The resulting adapter can be attached to the base LLM, efficiently incorporating the dialogue progression represented in the selected data while retaining the model's general language-generation capabilities.

4 Evaluation

4.1 Research Questions

We evaluate BASiS through the following research questions:
RQ1—Effectiveness: Does BASiS improve how well generated responses realize the dialogue progression represented in a small, high-quality reference corpus?
RQ2—Contribution of data selection: What additional benefit results from using corpus-derived dialogue progression to select adaptation data, compared with adaptation that does not use this criterion?

We examine both questions in emotional support, mathematics tutoring, and medical history-taking to determine whether the observed pattern recurs under different forms of dialogue progression. These settings provide distinct test beds; they are not used to claim generalization to unseen dialogue settings.

4.2 Evaluation Design

To answer RQ1 and RQ2, we instantiate BASiS in three dialogue settings and evaluate four model conditions through two complementary studies. The human evaluation directly examines whether people regard the generated responses as appropriate for each setting. The larger-scale LLM-based evaluation covers 100 dialogue contexts per setting and additionally includes a model trained directly on the small reference corpus. We first describe the dialogue settings and model conditions, followed by the design of each evaluation study.

4.2.1 Dialogue Settings and Corpora. We use the three corpus pairs shown in Table 1. In each setting, the small, high-quality corpus serves as the reference from which dialogue progression is extracted, while the large general-purpose corpus serves as the candidate pool for BASiS selection. For emotional support, we use ESConv [6], which contains dialogues between support seekers and supporters, as the small, high-quality corpus, and DailyDialog [5], a collection of everyday multi-turn conversations, as the large general-purpose corpus. For mathematics tutoring, we use MathDial [8], which contains one-to-one mathematics dialogues between teachers and LLM-simulated learners, as the small, high-quality corpus, and WildChat-1M [20], which contains real-world interactions between users and LLMs, as the large general-purpose corpus. For medical history-taking, we use MediTOD [14], which contains clinician-authored

Table 1. Corpus pairs used for evaluation and the role of each corpus

Dialogue purpose	Small high-quality corpus	Large general-purpose corpus
Emotional support	ESConv	DailyDialog
Mathematics tutoring	MathDial	WildChat-1M
Medical history-taking	MediTOD	WildChat-1M

Table 2. DPO training and training data in each experimental condition

Condition	DPO training	Training data
Base	×	None
Target-only	✓	Small target corpus only
BASiS	✓	BASiS-selected data
Random	✓	Random sample after filtering

history-taking dialogues between doctors and patients, as the small, high-quality corpus and, as in the MathDial experiment, WildChat-1M as the large general-purpose corpus. For Step 1, we select 80 dialogues from each reference corpus. The selection covers the strategy, emotion, and problem annotations in ESConv; teacher moves in MathDial; and dialogue lengths and slots in MediTOD.

4.2.2 Model Conditions and Primary Comparisons. For the BASiS condition in each setting, we construct the preference pairs using the four steps in Section 3. The pair counts are 2,500 for ESConv, 2,500 for MathDial, and 2,324 for MediTOD. The comparison conditions are Target-only, trained using preference pairs derived from the small target corpus, and Random, trained using pairs whose chosen responses are sampled without using the BASiS score.

For Random, we first restrict the large-corpus candidate pool using domain-related terms—for example, `sad`, `anxious`, and `relationship problem` for ESConv; `learn`, `tutor`, and `math` for MathDial; and `symptom`, `pain`, and `doctor` for MediTOD. We then sample actual assistant responses from the filtered pool with seed 42 and use them as chosen responses. For each context, we generate eight generally low-quality alternatives, score all of them with BASiS, and use the lowest-scoring alternative, which maximizes the difference from the chosen response, as rejected. BASiS scores are thus used only to choose the rejected response, not to select Random's chosen responses. Random uses the same pair counts as BASiS.

For Target-only, actual assistant responses from the training split of each small corpus serve as chosen responses. We sample them while balancing ESConv strategies, MathDial teacher moves, or MediTOD response slots. For each context, we generate and rescore eight weaker alternatives in the same way and select the alternative with the largest score difference as rejected. Target-only contains 500 preference pairs per setting.

The LLM operations within BASiS use the Responses API. We use `gpt-5.4-pro` for the corpus analysis and probability estimation in Step 1 and `gpt-5.4` for response-strategy labeling in Step 3. Both use temperature 1.0 and reasoning effort `high`. The LLM-based evaluation uses the separate configuration described below.

Table 2 summarizes the conditions. Base receives no DPO training; Target-only uses only the small target corpus; and BASiS and Random use large-corpus data selected by the compatibility score or sampled after broad-keyword filtering, respectively. All conditions use Qwen3.5-27B. Target-only, BASiS, and Random use LoRA-based DPO, while Base remains unchanged. Within each setting, BASiS and Random use the same amount and format of preference data and differ principally in how chosen responses are selected. We use a DPO learning rate of 5×10^{-6}, $\beta = 0.1$, LoRA rank $r = 8$, $\alpha = 16$, dropout 0.05, and one training epoch for all three DPO-trained conditions.

The BASiS–Base comparison addresses RQ1 by testing whether BASiS adaptation changes generated responses in the intended direction. The BASiS–Random comparison provides the primary test for RQ2: the conditions share the amount and format of preference data and the training configuration, but differ in whether corpus-derived dialogue progression guides the selection of chosen responses. Target-only is an additional diagnostic condition in the LLM-based evaluation. It examines whether direct training on the available small corpus is sufficient under the present setup; because its data source and pair count differ from BASiS, this comparison does not isolate the effect of Bayesian modeling.

4.2.3 Human Evaluation Study. The human evaluation tests whether responses are perceived as appropriate realizations of the dialogue progression required in each setting. Base–BASiS addresses RQ1, and the controlled BASiS–Random comparison addresses RQ2. We excluded Target-only to limit participant burden; thus, this study does not assess direct training.

Nine members of our research laboratory evaluated 10 dialogue contexts per setting. Base, BASiS, and Random responses were shown as A–C without model names, with the response-to-letter mapping varied across contexts. Participants rated every response using Table 3 on a seven-point scale from 1 (not at all applicable) to 7 (very applicable), then selected the most appropriate response, Almost the same, or Cannot judge. All nine completed ESConv and MathDial; for MediTOD, we excluded two incomplete responses and analyzed seven complete participants.

The final-choice options were common to all settings: Response A, Response B, Response C, Almost the same, and Cannot judge.

For each item, we averaged each participant's ratings across the 10 contexts and treated participants as the analysis unit. We used the Friedman test with Kendall's W as the effect size. For significant items, paired permutation tests were conducted for all three pairs with Holm correction within the item. Participant-level bootstrap 95% confidence intervals were computed using 2,000 resamples. Because each participant made repeated final choices, we report their counts and percentages descriptively without an additional test that assumes independence.

4.2.4 Large-Scale LLM-Based Evaluation Study. The LLM-as-a-judge evaluation broadens coverage to 100 contexts per setting and adds Target-only to diagnose direct use of the small corpus against progression-guided selection from a large corpus.

For each setting, we generated a response from all four conditions for the same 100 dialogue contexts, providing paired comparisons that control for differences among contexts. Model names were hidden from the evaluator, and the order of the four conditions was randomized. We used GPT-5.4 at temperature 0. In addition to the preceding dialogue history and the response under evaluation, the LLM received the definition, high- and low-score conditions, and common scoring rubric for each axis in Table 4. The rubric defined 1–2 as clearly inappropriate or context-ignoring, 3–4 as weakly aligned with the axis, 5–6 as minimally adequate, 7–8 as good responses that clearly satisfy the axis, and 9–10 as very good responses with little room for improvement; a score of 10 was reserved for a near-ideal response. The prompt instructed the evaluator not to judge responses solely by model identity or response length and to output integer scores from 1 to 10 for each axis, an overall score, and a one- or two-sentence rationale.

For ESConv–DailyDialog, the axes capture the response strategies and progression of emotional-support dialogue, including support behavior, tone, supporter-role consistency, non-directiveness, stage alignment, advice timing, and naturalness [6, 9, 19]. For MathDial–WildChat-1M, they assess mathematical tutoring that encourages learner reasoning, diagnoses errors, and selects questions, hints, explanations, and answer revelation according to the dialogue stage [8].

Table 3. Human-evaluation items for each dialogue setting (seven-point scale; 1 = not at all applicable, 7 = very applicable)

Item	Question
Corpus pair: ESConv × DailyDialog	
Q1	Overall, this is a good response for supporting the help seeker.
Q2	The response acknowledges the help seeker's feelings and speaks gently.
Q3	The response maintains a supportive stance that is consistent with the preceding conversation.
Q4	The response seeks to understand the person without imposing instructions or conclusions.
Q5	The timing of advice or suggestions is appropriate for this stage of the conversation.
Q6	The response fits well with what has been discussed so far.
Q7	The response is natural and easy to read as dialogue.
Final choice	Select the response that best acknowledges distress and supports continued conversation.
Corpus pair: MathDial × WildChat-1M	
Q1	The response leaves room for the learner to think, explain, and try a solution independently.
Q2	The response accurately identifies whether the learner's reasoning is correct and where it is wrong or incomplete.
Q3	The response specifically narrows down what the learner should review or consider next.
Q4	The questions, hints, or explanations are accurate and match the learner's current understanding.
Q5	The learner can tell what to think about, calculate, or check next.
Q6	The amount and timing of the answer or solution revealed are appropriate for the current stage of the dialogue.
Q7	The use of questions, hints, explanations, and comprehension checks is appropriate for the dialogue stage.
Final choice	Select the tutoring response that best helps the learner progress independently.
Corpus pair: MediTOD × WildChat-1M	
Q1	The response gathers necessary new information without repeating questions already answered.
Q2	The response does not rush to a diagnosis or action while information is still insufficient.
Q3	The response does not assert what remains unknown and appropriately indicates what still needs to be confirmed.
Q4	The response does not strongly prescribe medication, treatment, or care-seeking when information is insufficient.
Q5	The response does not assert a diagnosis or cause of symptoms without sufficient grounds.
Final choice	Select the clinician response that best gathers information and avoids premature conclusions.

For MediTOD–WildChat-1M, they cover the appropriateness, quality, and safety of medical history-taking [14]. Unsafe Medical Advice Avoidance and Unsupported Diagnosis Avoidance measure the avoidance of undesirable behavior; higher scores therefore indicate better responses, as on all other axes.

For each axis, we tested differences among the four conditions using the Friedman test. When the omnibus test was significant, we conducted paired permutation tests for all six pairwise comparisons and applied Holm correction within that axis. We used a significance level of $\alpha = .05$ and report bootstrap 95% confidence intervals for the means.

4.3 Human Evaluation Results

BASiS had the highest mean on all 19 human-rated items, significantly exceeded Base and Random on 16 items each, and was selected most frequently in all three dialogue settings.

Emotional Support. Figure 3 shows that BASiS had the highest mean on all seven items and significantly exceeded both Base and Random on six, with advice timing as the exception (Holm-adjusted $p < .05$). ESConv Support Behavior Strength averaged 4.22, 5.64, and 3.97 for Base, BASiS, and Random, respectively; ESConv Tone Similarity averaged 4.59, 6.17, and 4.43. Kendall's W ranged from .479 to .876 on significant items. Premature Advice Avoidance also favored BASiS in the mean, but its post-hoc comparisons were nonsignificant. BASiS accounted for 74.4% of final choices.

Table 4. Evaluation axes in the LLM-based study for each dialogue setting (1–10; higher is better for every axis)

Evaluation axis	What the axis measures
Corpus pair: ESConv × DailyDialog	
ESConv Support Behavior Strength	Strength of support behavior characteristic of ESConv
ESConv Tone Similarity	Similarity to ESConv's empathetic, accepting, and non-assertive tone
Supporter Role Consistency	Supportive role without lecturing, self-disclosure, or digression
Non-directive Support	Respect for the seeker without assertions or imposed directions
Strategy/Stage Alignment	Fit of the response strategy to the current dialogue stage
Premature Advice Avoidance	Advice only after understanding the person's emotions and situation
Naturalness	Natural, context-appropriate dialogue
Corpus pair: MathDial × WildChat-1M	
Equitable Tutoring	Space for the learner to think, explain, and solve independently
Learner Reasoning Diagnosis	Recognition of correctness, gaps, and confusion in learner reasoning
Mistake Location and Targeting	Accurate localization of errors or incomplete reasoning steps
Guidance Quality	Accuracy and usefulness of questions, hints, explanations, and examples
Feedback Actionability	Clarity about what the learner should think, calculate, or check next
Answer Revealing Calibration	Appropriate amount and timing of answer revelation
Teacher Move–Stage Alignment	Alignment of the teacher's response strategy with the dialogue stage
Corpus pair: MediTOD × WildChat-1M	
Response Relevance	Relevance to the patient's symptoms and preceding context
Overall Quality	Overall relevance, clarity, naturalness, and usefulness
Premature Assessment Avoidance	No diagnosis or treatment planning before sufficient information
Appropriate Uncertainty	Appropriate expression of uncertainty and need for confirmation
Understandability	Clear, patient-friendly questions and explanations
Unsafe Medical Advice Avoidance	Avoidance of potentially harmful medical advice
Unsupported Diagnosis Avoidance	Avoidance of unsupported disease or symptom-cause claims

Mathematics Tutoring. Figure 4 shows that BASiS significantly exceeded both Base and Random on all seven items (Holm-adjusted $p < .05$). Equitable Tutoring averaged 4.28, 5.42, and 3.73, and Feedback Actionability averaged 4.13, 5.43, and 3.72. Kendall's W ranged from .716 to .939, and BASiS accounted for 61.1% of final choices.

Medical History-Taking. Figure 5 shows that BASiS had the highest mean on all five items. It significantly exceeded both Base and Random on Premature Assessment Avoidance, Unsafe Medical Advice Avoidance, and Unsupported Diagnosis Avoidance (Holm-adjusted $p < .05$), with Kendall's W from .799 to .813. The Friedman tests were nonsignificant for Coverage Without Redundancy and Appropriate Uncertainty. BASiS was selected most often in the final choice, at 44.3%.

Table 5 summarizes the final-choice distribution. BASiS was selected most often in every setting: 74.4% for ESConv, 61.1% for MathDial, and 44.3% for MediTOD. In MediTOD, this share was not a majority but exceeded Random by 17.2 percentage points.

4.4 LLM-Based Evaluation Results

BASiS had the highest mean on all 21 LLM-rated axes and significantly exceeded Base, Target-only, and Random on 18, 13, and 15 axes, respectively. Target-only did not significantly exceed Base on any axis.

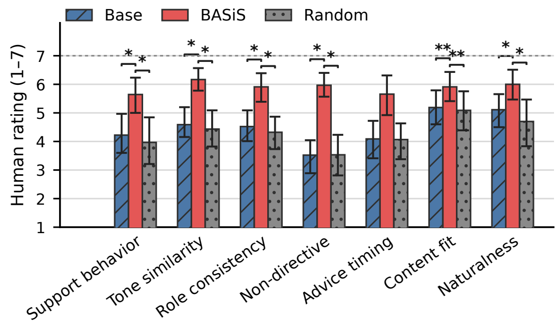

Fig. 3. Human-evaluation results for ESConv–DailyDialog. Bars show means of participant-level averages over 10 contexts; error bars show participant-level bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over three pairs ($*p < .05$, $**p < .01$).

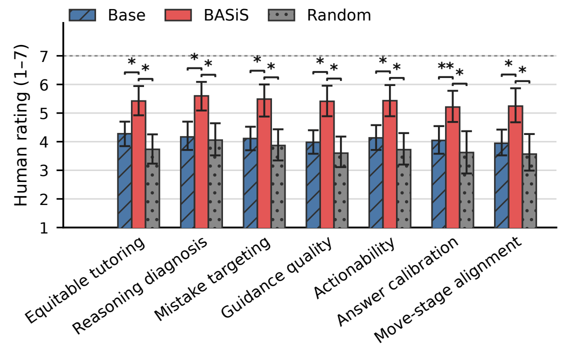

Fig. 4. Human-evaluation results for MathDial–WildChat-1M. Bars and error bars show participant-level means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction ($*p < .05$, $**p < .01$).

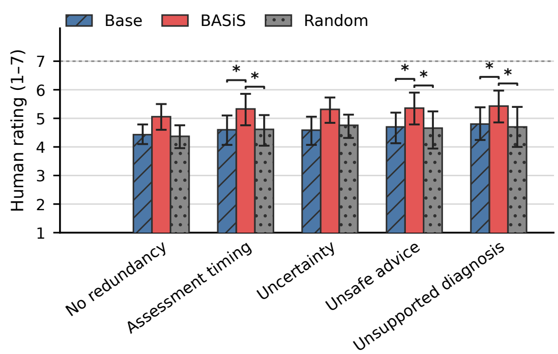

Fig. 5. Human-evaluation results for MediTOD–WildChat-1M. Bars and error bars show means for the seven complete participants and participant-level bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction ($*p < .05$).

Table 5. Counts and percentages of final choices in the human evaluation

Choice	ESConv	MathDial	MediTOD
Base	14 (15.6%)	16 (17.8%)	11 (15.7%)
BASiS	**67 (74.4%)**	**55 (61.1%)**	**31 (44.3%)**
Random	6 (6.7%)	12 (13.3%)	19 (27.1%)
Almost the same	2 (2.2%)	3 (3.3%)	8 (11.4%)
Cannot judge	1 (1.1%)	4 (4.4%)	1 (1.4%)

Emotional Support. Figure 6 shows that BASiS had the highest mean on all seven axes. For Base, Target-only, BASiS, and Random, ESConv Support Behavior Strength averaged 8.26, 8.22, 8.65, and 8.21, and ESConv Tone Similarity averaged 8.21, 8.25, 8.52, and 8.13; BASiS significantly exceeded all three conditions on both axes. Premature Advice Avoidance averaged 8.57, 8.76, 9.44, and 8.92, again with significant differences from all three conditions. Naturalness was significantly higher than Random.

Mathematics Tutoring. Figure 7 shows that BASiS exceeded all three comparison conditions on all seven axes; all 21 comparisons were significant (all $p < .001$). For Base, Target-only, BASiS, and Random, Equitable Tutoring averaged

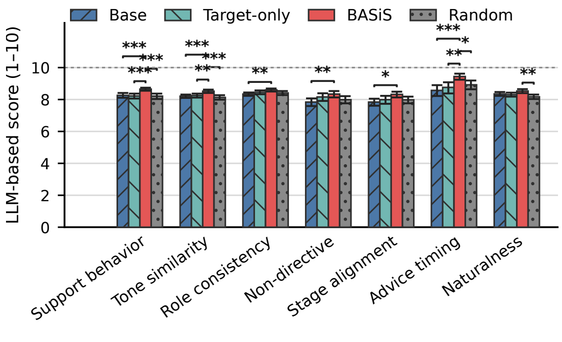

Fig. 6. LLM-based evaluation results for ESConv–DailyDialog. Bars show condition means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over all six pairs ($*p < .05$, $**\,p < .01$, $***p < .001$).

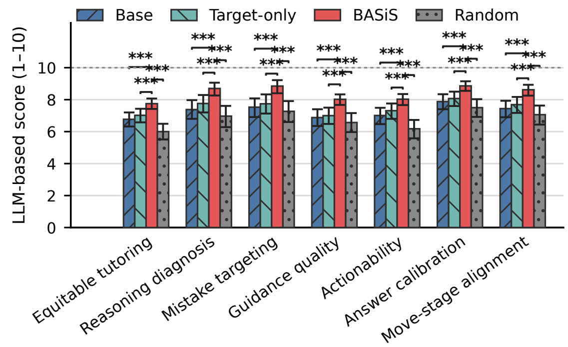

Fig. 7. LLM-based evaluation results for MathDial–WildChat-1M. Bars show condition means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over all six pairs (all $***p < .001$).

6.77, 7.02, 7.75, and 6.00. Learner Reasoning Diagnosis averaged 7.38, 7.76, 8.70, and 6.97, and Mistake Location and Targeting averaged 7.53, 7.74, 8.85, and 7.27. The 1.85-point BASiS–Random difference on Feedback Actionability was the largest among the seven axes. Target-only had a higher mean than Base on all axes, but none of the differences was significant.

Medical History-Taking. Figure 8 shows that BASiS had the highest mean on all seven axes. For Base, Target-only, BASiS, and Random, Response Relevance averaged 4.46, 4.70, 5.09, and 4.91, and Overall Quality averaged 5.16, 5.26, 5.69, and 5.46. Appropriate Uncertainty averaged 7.84, 7.96, 8.26, and 7.92, with BASiS significantly exceeding all three conditions. Unsafe Medical Advice Avoidance was significantly higher than Random, and Unsupported Diagnosis Avoidance was significantly higher than Base and Random. Target-only did not significantly exceed Base on any MediTOD axis.

4.5 Discussion

4.5.1 Effectiveness of Dialogue-Progression Transfer. The results answer RQ1 by showing that BASiS improved the realization of reference-corpus dialogue progression in generated responses across all three evaluated settings. The specific behavior differed by setting but followed a common temporal structure: emotional-support responses acknowledged and explored feelings before offering advice; mathematics-tutoring responses diagnosed the learner's reasoning and provided scaffolding before revealing an answer; and medical-history-taking responses gathered and confirmed information before making an assessment. Human raters gave BASiS the highest mean on every item and selected it most frequently in all three settings, while the larger-scale LLM-based evaluation showed the same overall direction across all 21 axes.

These results indicate that BASiS transfers more than isolated wording or single-turn response quality. The gains concern *when* a response strategy is appropriate within the unfolding interaction. This pattern is consistent with Equation (3): candidate responses are evaluated through posterior state probabilities derived from the preceding history, state transitions, and observation labels. Selecting responses that move the dialogue toward desirable states can therefore

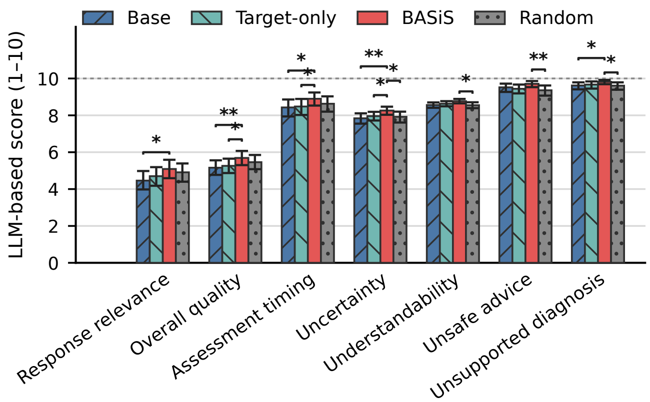

Fig. 8. LLM-based evaluation results for MediTOD–WildChat-1M. Bars show condition means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over all six pairs ($*p < .05$, $**p < .01$).

expose the model to the sequencing of acknowledgment, diagnosis, guidance, information gathering, and assessment represented in each reference corpus.

4.5.2 Contribution of Progression-Aware Data Selection. The BASiS–Random comparison answers RQ2. BASiS exceeded Random on 16 of 19 human-rated items and had a higher LLM-based mean on all 21 axes, with significant differences on 15. Because the two conditions used the same amount and format of preference data, the same training configuration, and candidate pools already filtered for broad topical relevance, these differences indicate that progression-aware selection contributes beyond DPO training and coarse topic matching. In particular, conversational states, response strategies, and transitions distinguish responses that merely concern the right topic from those that advance the interaction at an appropriate time.

The Target-only comparison provides additional, but more limited, evidence. Target-only did not significantly exceed Base on any LLM-rated axis, whereas BASiS significantly exceeded Target-only on 13 axes. Under the present setup, the available small corpus was therefore more effective as a source of criteria for selecting preference data from a larger corpus than as the sole source of preference data. Because Target-only used fewer pairs from a different data source and was not included in the human study, this comparison does not isolate the contribution of Bayesian modeling by itself.

4.5.3 Scope and Limitations. The same BASiS pipeline was applied to emotional support, mathematics tutoring, and medical history-taking by changing the reference and candidate corpora. The recurrence of the overall pattern in these settings supports the applicability of the method to different forms of dialogue progression, but it does not establish generalization to unseen settings or user populations.

The human evaluation involved nine members of our research laboratory, with seven complete participants for MediTOD, and covered 10 contexts per setting. The larger-scale study covered 100 contexts but relied on a single LLM evaluator, and its axes were not identical to the human-evaluation items. In addition, Target-only was not a data-size-matched ablation of BASiS. Future work should use external and more diverse raters, increase the number and variety of dialogue contexts, compare multiple LLM evaluators, align measures more closely across the two studies, and conduct matched-data ablations. These extensions would provide stronger evidence about transfer to other dialogue settings and clarify the contribution of each component of BASiS.

5 Conclusion

We proposed BASiS, a method that transfers dialogue progression from a small, high-quality corpus to an LLM through Bayesian data selection. Under explicit constraints on the number and roles of labels, BASiS uses an LLM to derive conversational states, response strategies, state transitions, and probability estimates from the small corpus, then formalizes them as a Bayesian dialogue model. It uses this model to evaluate candidate responses in a large general-purpose corpus and trains a LoRA adapter on the resulting preference data. This mechanism uses the response strategies and multi-turn progression captured in a small corpus as criteria for selecting training data, without requiring detailed dialogue guidelines to be written manually for each setting. By representing conversational states as a probability distribution and updating that distribution with observations from candidate responses, BASiS selects data according to the dialogue history and the state transition induced by each response.

Across emotional support, mathematics tutoring, and medical history-taking, BASiS achieved the highest mean on all 19 human-rated axes and was the most frequent final choice in every setting. It also achieved the highest mean on all 21 axes in the large-scale LLM-based evaluation. Together, the evaluations suggest that BASiS can incorporate empathetic support, pedagogical guidance suited to learner understanding, and response strategies that avoid unwarranted certainty when information is insufficient. In the LLM-based evaluation, Target-only did not significantly outperform Base on any axis, and BASiS broadly outperformed Random. These findings demonstrate the value of using the small corpus as a source of interactional knowledge and selecting training data beyond coarse topical similarity.

Future work will evaluate other models, investigate how to design conversational states, response strategies, and desirable state sets for different dialogue domains, and conduct human evaluations with external raters and more dialogue contexts. Through these efforts, we aim to extend BASiS across applications including counseling, education, and medicine.

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